



Department of Computer Science & Engineering
JD College of Engineering and Management, Nagpur - 441501
(An Autonomous Institute, with NAAC “A” Grade)

Report on

Title: Rocket Launch Animation using Python

Submitted by :

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Class: 5th sem, CSE-A

Sub : Interactive Graphics

Introduction

This project demonstrates how Python can be used to create an interactive and visually appealing animation. Using the Pygame library, I designed a simulation of a rocket launch that includes a countdown, background stars, realistic flames, and smoke effects.

Objectives

- To understand the use of Pygame for creating graphics and animations.
- To implement shapes, colors, and motion in a real-time simulation.
- To design a step-by-step rocket launch with countdown, upward movement, and background effects.

Methodology

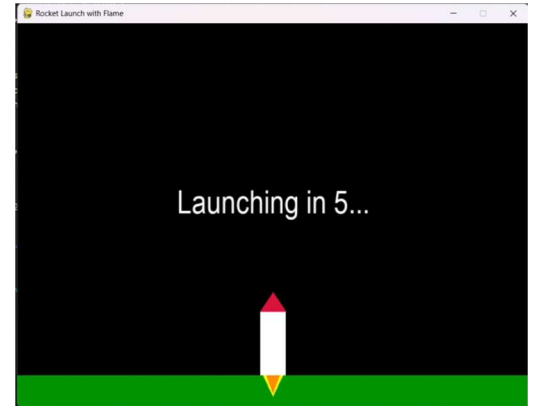
1. Modules Used:

- Pygame for graphics and animation.

- Sys for program exit.
- Time for countdown delays.
- Random for generating stars, flames, and smoke variations.

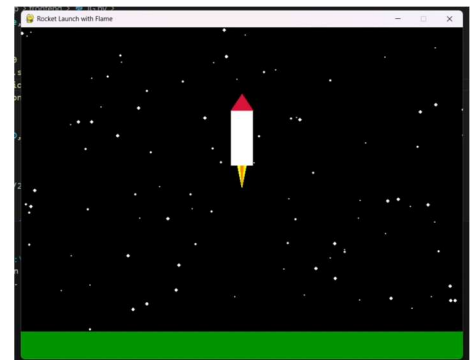
2. Implementation:

- Defined screen dimensions and colors.
- Created rocket structure using rectangles, polygons, and color layers.
- Generated two hundred random stars for the background.
- Added countdown text before launch.
- Animated the rocket moving upward with flames and smoke trails.
- Updated the screen at sixty frames per second for smooth animation.



Features of the Program :

- Countdown from 5 to 1 before liftoff.
- Rocket body with red stripes, grey nose cone, and boosters.
- Flames in red, orange, and yellow with flickering effect.
- Moving stars with twinkling animation.
- Realistic smoke trail that expands and fades.
- Smooth upward motion until the rocket leaves the screen



Conclusion : This project uses Python and Pygame to create a rocket launch animation, giving me practical experience with graphics and events.”